

From Grover Search to Quantum Walks

Search under structure



Rupanshu Shah

Entry No.: 2025CSZ8228

Supervisor: **Prof. Rajendra Kumar**

Indian Institute of Technology Delhi

1. Unstructured search

Problem

Given oracle access to

$$f : [N] \rightarrow \{0, 1\},$$

find an $x \in [N]$ such that $f(x) = 1$.

1. Unstructured search

Problem

Given oracle access to

$$f : [N] \rightarrow \{0, 1\},$$

find an $x \in [N]$ such that $f(x) = 1$.

Classical query lower bound

If there is a unique marked item, randomized classical algorithms require

$$\Omega(N)$$

queries for constant success probability.

2. Grover's search: the quantum benchmark

Same problem

Given $f : [N] \rightarrow \{0, 1\}$, find x such that $f(x) = 1$.

2. Grover's search: the quantum benchmark

Same problem

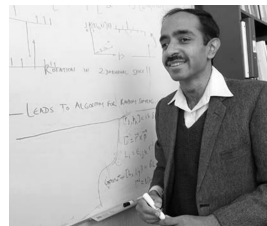
Given $f : [N] \rightarrow \{0, 1\}$, find x such that $f(x) = 1$.

Quantum query complexity

Grover's algorithm solves this using

$$O(\sqrt{N})$$

queries, and this is optimal up to constants.



Lov K. Grover

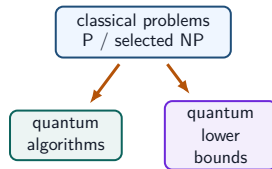
B.Tech. Electrical Engineering, IIT
Delhi, 1981

Conferred IIT Delhi Distinguished
Alumnus Award in 2021

3. Research lens: classical problems, quantum viewpoint

High-level PhD direction

Study classical computational problems through the quantum lens: primarily problems in P, and eventually selected NP-type problems.



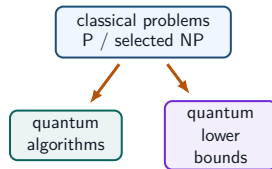
3. Research lens: classical problems, quantum viewpoint

High-level PhD direction

Study classical computational problems through the quantum lens: primarily problems in P, and eventually selected NP-type problems.

Two complementary goals

- ▶ design the best possible quantum algorithms;
- ▶ prove quantum lower bounds / fine-grained limitations.



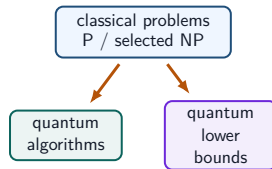
3. Research lens: classical problems, quantum viewpoint

High-level PhD direction

Study classical computational problems through the quantum lens: primarily problems in P, and eventually selected NP-type problems.

Two complementary goals

- ▶ design the best possible quantum algorithms;
- ▶ prove quantum lower bounds / fine-grained limitations.



This talk focuses on the algorithmic side.

4. Why Grover-like speedups matter

Popular picture

A large fault-tolerant quantum computer would threaten widely used public-key cryptosystems via Shor-type algorithms.

4. Why Grover-like speedups matter

Popular picture

A large fault-tolerant quantum computer would threaten widely used public-key cryptosystems via Shor-type algorithms.

Algorithmic picture

Grover-like quadratic speedups target search, optimization, and verification primitives that appear across many classical algorithms.

4. Why Grover-like speedups matter

Popular picture

A large fault-tolerant quantum computer would threaten widely used public-key cryptosystems via Shor-type algorithms.

Algorithmic picture

Grover-like quadratic speedups target search, optimization, and verification primitives that appear across many classical algorithms.

crypto-breaking
narrative

vs.

search/optimization
primitives

Motivation here: understand where such algorithmic speedups can actually be proved.

5. What changes when there is structure?

Unstructured search

Any item can be queried next. Geometry is irrelevant.

$[N]$ is just a set.

5. What changes when there is structure?

Unstructured search

Any item can be queried next. Geometry is irrelevant.

$[N]$ is just a set.

Structured search

The candidates may have geometry, constraints, or recursive organization.

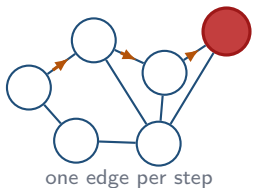


marked object hidden inside structure

6. Two algorithmic responses to structure

1. Random walks

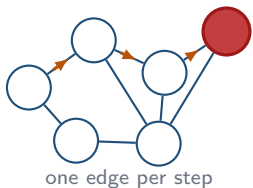
Use local moves on an explicit or implicit state space.



6. Two algorithmic responses to structure

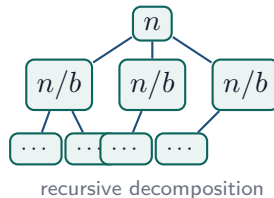
1. Random walks

Use local moves on an explicit or implicit state space.



2. Divide and conquer

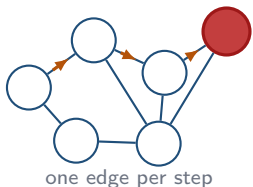
Use recursive structure: solve subproblems, then combine.



6. Two algorithmic responses to structure

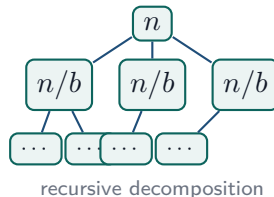
1. Random walks

Use local moves on an explicit or implicit state space.



2. Divide and conquer

Use recursive structure: solve subproblems, then combine.



Main thread: quantum walks.

Future direction: quantum divide and conquer.

7. Graph search: marked vertex problem

Problem

Given a graph $G = (V, E)$ and a marking oracle

$$f : V \rightarrow \{0, 1\}, \quad \mathcal{M} = \{v \in V : f(v) = 1\},$$

find some $v \in \mathcal{M}$.

7. Graph search: marked vertex problem

Problem

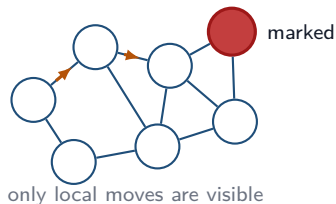
Given a graph $G = (V, E)$ and a marking oracle

$$f : V \rightarrow \{0, 1\}, \quad \mathcal{M} = \{v \in V : f(v) = 1\},$$

find some $v \in \mathcal{M}$.

Local-access model

At vertex v , the algorithm can check whether v is marked and move to a neighbour of v .



7. Graph search: marked vertex problem

Problem

Given a graph $G = (V, E)$ and a marking oracle

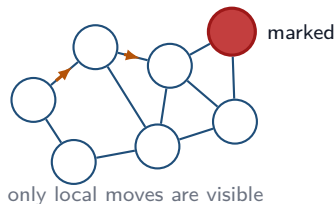
$$f : V \rightarrow \{0, 1\}, \quad \mathcal{M} = \{v \in V : f(v) = 1\},$$

find some $v \in \mathcal{M}$.

Local-access model

At vertex v , the algorithm can check whether v is marked and move to a neighbour of v .

Complete graph $\Rightarrow$ ordinary Grover search.

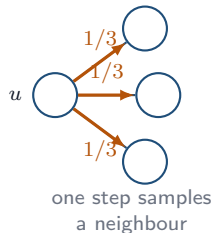


8. Classical random walk on a graph

Transition matrix

For an undirected graph $G = (V, E)$,

$$P_{u,v} = \begin{cases} \frac{1}{\deg(u)}, & (u,v) \in E, \\ 0, & \text{otherwise.} \end{cases}$$

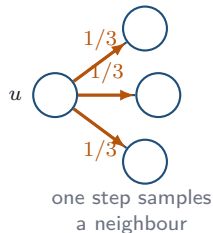


8. Classical random walk on a graph

Transition matrix

For an undirected graph $G = (V, E)$,

$$P_{u,v} = \begin{cases} \frac{1}{\deg(u)}, & (u,v) \in E, \\ 0, & \text{otherwise.} \end{cases}$$



Distribution update

For a row distribution vector p_t ,

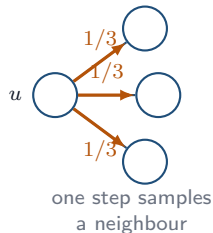
$$p_{t+1} = p_t P, \quad p_t = p_0 P^t.$$

8. Classical random walk on a graph

Transition matrix

For an undirected graph $G = (V, E)$,

$$P_{u,v} = \begin{cases} \frac{1}{\deg(u)}, & (u,v) \in E, \\ 0, & \text{otherwise.} \end{cases}$$



Distribution update

For a row distribution vector p_t ,

$$p_{t+1} = p_t P, \quad p_t = p_0 P^t.$$

Why even do a random walk?

9. Why even do a random walk?

Reason

Often the relevant state space is too large, too implicit, or too structured to scan globally.

9. Why even do a random walk?

Reason

Often the relevant state space is too large, too implicit, or too structured to scan globally.

When the graph is given

- ▶ local neighbour queries;
- ▶ small working memory;
- ▶ BFS stores large layers;
- ▶ DFS needs visited-state memory.

9. Why even do a random walk?

Reason

Often the relevant state space is too large, too implicit, or too structured to scan globally.

When the graph is given

- ▶ local neighbour queries;
- ▶ small working memory;
- ▶ BFS stores large layers;
- ▶ DFS needs visited-state memory.

When the graph is designed

- ▶ vertices are partial solutions;
- ▶ edges are local updates;
- ▶ marked vertices encode certificates;
- ▶ the graph is only implicit.

9. Why even do a random walk?

Reason

Often the relevant state space is too large, too implicit, or too structured to scan globally.

When the graph is given

- ▶ local neighbour queries;
- ▶ small working memory;
- ▶ BFS stores large layers;
- ▶ DFS needs visited-state memory.

When the graph is designed

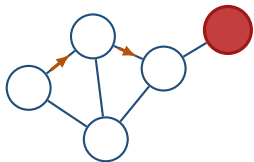
- ▶ vertices are partial solutions;
- ▶ edges are local updates;
- ▶ marked vertices encode certificates;
- ▶ the graph is only implicit.

Examples: networks and web graphs; subset spaces, configuration spaces, and algorithm-designed state spaces.

10. Quantum walks: the graph need not be input

Given graph

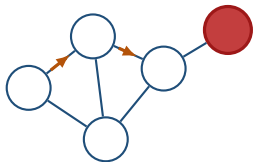
A quantum walk can search an explicitly given graph under locality constraints.



10. Quantum walks: the graph need not be input

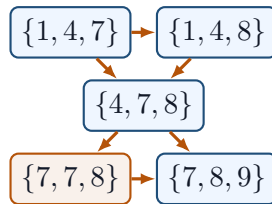
Given graph

A quantum walk can search an explicitly given graph under locality constraints.



Constructed graph

For many algorithmic problems, we build a graph over states such as subsets, paths, assignments, or intervals.

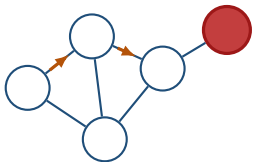


implicit subset graph

10. Quantum walks: the graph need not be input

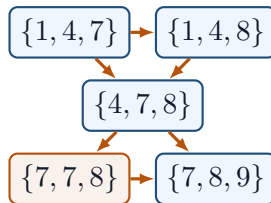
Given graph

A quantum walk can search an explicitly given graph under locality constraints.



Constructed graph

For many algorithmic problems, we build a graph over states such as subsets, paths, assignments, or intervals.



implicit subset graph

Quantum walks are an algorithm-design framework, not only a tool for huge input graphs.

11. Example 1: glued trees

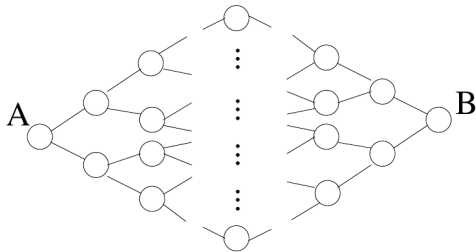


Figure adapted from Ambainis' survey, reproducing the Childs-Farhi-Gutmann glued-trees graph.

11. Example 1: glued trees

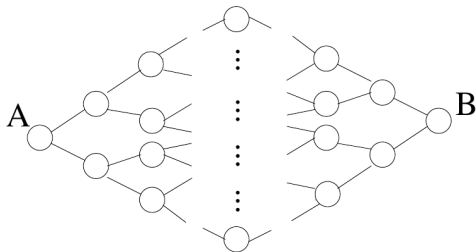


Figure adapted from Ambainis' survey, reproducing the Childs-Farhi-Gutmann glued-trees graph.

Classical random walk

Expected hitting time from A to B :

$$\Omega(2^d).$$

11. Example 1: glued trees

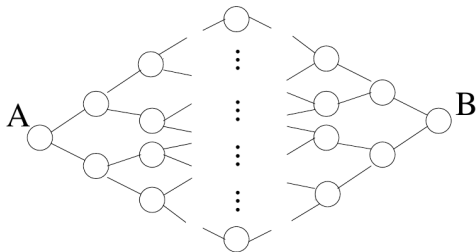


Figure adapted from Ambainis' survey, reproducing the Childs–Farhi–Gutmann glued-trees graph.

Classical random walk

Expected hitting time from A to B :

$$\Omega(2^d).$$

Continuous-time quantum walk

Constant success probability in

$$O(d^2) \text{ queries.}$$

12. Example 2: element distinctness

Problem

Given $x_1, x_2, \dots, x_N \in [M]$, decide whether

$$\exists i \neq j \quad x_i = x_j.$$

12. Example 2: element distinctness

Problem

Given $x_1, x_2, \dots, x_N \in [M]$, decide whether

$$\exists i \neq j \quad x_i = x_j.$$

Classical query lower bound

Element distinctness requires

$$\Omega(N)$$

queries classically.

12. Example 2: element distinctness

Problem

Given $x_1, x_2, \dots, x_N \in [M]$, decide whether

$$\exists i \neq j \quad x_i = x_j.$$

Classical query lower bound

Element distinctness requires

$$\Omega(N)$$

queries classically.

Quantum-walk algorithm

Ambainis' optimal quantum query algorithm:

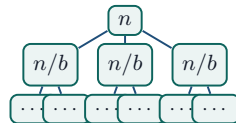
$$O(N^{2/3}).$$

There is no input graph. The algorithm constructs a walk over subsets of indices.

Divide and conquer

Classical recurrence

$$C(n) \leq a C(n/b) + C_{\text{aux}}(n).$$



Classical: inspect a
recursive branches

Divide and conquer

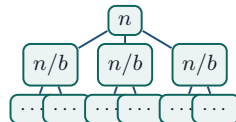
Classical recurrence

$$C(n) \leq a C(n/b) + C_{\text{aux}}(n).$$

Quantum recurrence when composition works

$$C_Q(n) \leq \sqrt{a} C_Q(n/b) + C_{\text{aux}}^Q(n).$$

branch factor changes: $a \rightsquigarrow \sqrt{a}$



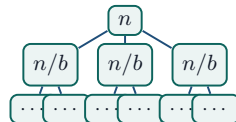
Classical: inspect a
recursive branches

Quantum: search
over branches

Divide and conquer

Classical recurrence

$$C(n) \leq a C(n/b) + C_{\text{aux}}(n).$$



Quantum recurrence when composition works

$$C_Q(n) \leq \sqrt{a} C_Q(n/b) + C_{\text{aux}}^Q(n).$$

Classical: inspect a
recursive branches

Quantum: search
over branches

branch factor changes: $a \rightsquigarrow \sqrt{a}$

Why this is nontrivial

Recursive quantum subroutines must be called coherently inside superpositions, and their errors must remain controlled across levels.

14. What exactly remains to be explored?

Existing toolkit

- ▶ Grover / amplitude amplification variants;
- ▶ quantum walks for marked-state search;
- ▶ recent quantum divide-and-conquer frameworks.

14. What exactly remains to be explored?

Existing toolkit	Research task
▶ Grover / amplitude amplification variants; ▶ quantum walks for marked-state search; ▶ recent quantum divide-and-conquer frameworks.	For concrete classical problems, identify the right state space, diffusion/walk operator, and recursive combiner that make these tools applicable.

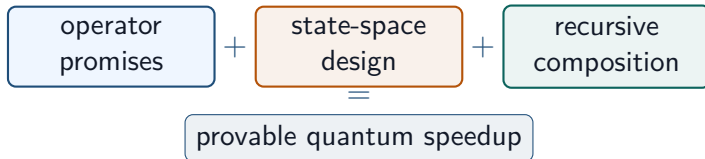
14. What exactly remains to be explored?

Existing toolkit

- ▶ Grover / amplitude amplification variants;
- ▶ quantum walks for marked-state search;
- ▶ recent quantum divide-and-conquer frameworks.

Research task

For concrete classical problems, identify the right state space, diffusion/walk operator, and recursive combiner that make these tools applicable.



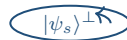
15. Future research directions

1. Use relaxed Grover-like conditions as a tool.

- ▶ identify which problem-derived operators satisfy usable spectral promises;
- ▶ quantify the overhead when diffusion is not exact Grover diffusion.

Relaxed diffusion condition


$$U |\psi_s\rangle = |\psi_s\rangle$$


$$|\psi_s\rangle^\perp$$

orthogonal eigenspaces
receive nonzero phases

15. Future research directions

1. Use relaxed Grover-like conditions as a tool.

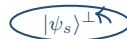
- ▶ identify which problem-derived operators satisfy usable spectral promises;
- ▶ quantify the overhead when diffusion is not exact Grover diffusion.

2. Extend quantum divide and conquer.

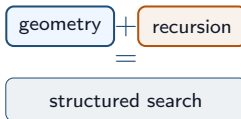
- ▶ combining functions beyond AND/OR and SWITCH-CASE;
- ▶ understand when super-quadratic speedups are possible.

Relaxed diffusion condition


$$U|\psi_s\rangle = |\psi_s\rangle$$


$$|\psi_s\rangle^\perp$$

orthogonal eigenspaces
receive nonzero phases



15. Future research directions

1. Use relaxed Grover-like conditions as a tool.

- ▶ identify which problem-derived operators satisfy usable spectral promises;
- ▶ quantify the overhead when diffusion is not exact Grover diffusion.

2. Extend quantum divide and conquer.

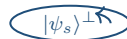
- ▶ combining functions beyond AND/OR and SWITCH-CASE;
- ▶ understand when super-quadratic speedups are possible.

3. Pair algorithms with lower bounds.

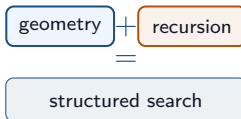
- ▶ fine-grained quantum limits for problems in P.

Relaxed diffusion condition


$$U|\psi_s\rangle = |\psi_s\rangle$$


$$|\psi_s\rangle^\perp$$

orthogonal eigenspaces
receive nonzero phases



Thank you

Questions?

Quantum walks: search when structure is algorithmically useful.

Appendix: what is the walk in element distinctness?

Johnson-graph viewpoint

A vertex is a subset $S \subseteq [N]$ of fixed size r . Two subsets are adjacent if they differ in one index.

Ambainis, Quantum walk algorithm for element distinctness; summarized in Ambainis' 2003 survey.

Appendix: what is the walk in element distinctness?

Johnson-graph viewpoint

A vertex is a subset $S \subseteq [N]$ of fixed size r . Two subsets are adjacent if they differ in one index.

Marked subset

S is marked if it already contains a collision:

$$\exists i, j \in S, i \neq j \quad x_i = x_j.$$

Ambainis, Quantum walk algorithm for element distinctness; summarized in Ambainis' 2003 survey.

Appendix: what is the walk in element distinctness?

Johnson-graph viewpoint

A vertex is a subset $S \subseteq [N]$ of fixed size r . Two subsets are adjacent if they differ in one index.

Marked subset

S is marked if it already contains a collision:

$$\exists i, j \in S, i \neq j \quad x_i = x_j.$$

Why this is useful

Checking uses stored values inside S ; updating to a neighbour changes one index and costs one new query.

Ambainis, Quantum walk algorithm for element distinctness; summarized in Ambainis' 2003 survey.

References used in the deck

- ▶ L. K. Grover, "A fast quantum mechanical algorithm for database search," STOC 1996.
- ▶ IIT Delhi Alumni Affairs, Distinguished Alumnus profile: Lov K. Grover, award year 2021.
- ▶ A. Ambainis, "Quantum Walks and Their Algorithmic Applications," IJQI 2003.
- ▶ A. Childs, E. Farhi, S. Gutmann, "An example of the difference between quantum and classical random walks," QIP 2002.
- ▶ A. Childs, R. Cleve, E. Deotto, E. Farhi, S. Gutmann, D. Spielman, "Exponential algorithmic speedup by quantum walk," STOC 2003.
- ▶ A. Ambainis, "Quantum walk algorithm for element distinctness," arXiv:quant-ph/0311001.
- ▶ Y. Shi, "Quantum lower bounds for the collision and the element distinctness problems," FOCS 2002.
- ▶ A. Childs, R. Kothari, M. Kovacs-Deak, A. Sundaram, D. Wang, "Quantum divide and conquer," arXiv:2210.06419.
- ▶ J. Allcock, J. Bao, A. Belovs, T. Lee, M. Santha, "On the quantum time complexity of divide and conquer," arXiv:2311.16401.